Problem Solving Case #1 (25 points)

You are assessing a bond position in a fixed-income portfolio. The bond details are as follows: a 10-year, 7.5% semiannual-pay corporate bond maturing on February 15, 2030. The current date is October 20, 2026, and the bond's YTM is 10%, quoted on a street-convention (30/360 day count) semiannual bond basis.

1. Using the provided information and the estimation method, calculate the modified duration of the bond position for a 25-bps yield change. (5 marks)
   using the 30/360 day-count convention:

   accrued days (Aug 15 - Oct 20, 2026): $(10-8)(30)+(20-15)=65$ days

   fraction of period left to next coupon (Feb 15, 2027): $f=\frac{180-65}{180}=0.6389$

   remaining payments (Feb 2027, Aug 2027, ..., Feb 2030): 7

   (YTM=10%, i.e. 5% per half-year):

   value at next coupon date: $3.75\left(1+\frac{1-1.05^{-6}}{0.05}\right)+100(1.05)^{-6}=\$97.41$

   dirty (full) price: $\frac{97.41}{(1.05)^{0.6389}}=\$94.42$

   accrued interest: $\left(\frac{65}{180}\right)(3.75)=\$1.35$

   clean price: $V_0=\$94.42-\$1.35=\$93.06$

   (-25 bp, YTM=9.75%): $V_-=95.08$ dirty, 93.73 clean

   (+25 bp, YTM=10.25%): $V_+=93.76$ dirty, 92.40 clean

   modified duration $\approx\frac{V_--V_+}{2\Delta y V_0}=\frac{95.08-93.76}{2\times0.0025\times94.42}\approx 2.81$ years The denominator must be the **full** (dirty) price 94.42: accrued interest does not react to the yield, so the position's sensitivity is driven by the full price, and the answer is **modified duration ≈ 2.81 years**. (Dividing by the clean price 93.06 gives 2.85; that variant understates the sensitivity because the accrued interest is part of the value at risk.)

2. Define the concepts of Macaulay duration and modified duration. How are they related? Using the bond's modified duration, compute its Macaulay duration. (4 marks)
   Macaulay duration: the present-value-weighted average time (in years) until the bond's cash flows are received, $\text{MacDur}=\frac{\sum_t t\cdot \text{PV}(\text{CF}_t)}{V_0}$.
   Modified duration: the percentage price change per unit (100%) change in yield, $\text{ModDur}=-\left(\frac{1}{V_0}\right)\frac{\mathrm{d}V}{\mathrm{d}y}$.
   Relationship: $\text{ModDur}=\frac{\text{MacDur}}{1+\frac{\text{YTM}}{k}}\iff \text{MacDur}=\text{ModDur}\left(1+\frac{\text{YTM}}{k}\right)$, where k is the number of coupon payments per year (here $k=2$).
   $\text{MacDur}=2.8056(1.05)=2.95$ years (built on the full-price modified duration) (The clean-price variant gives 2.8464 × 1.05 = 2.99 years; 2.95 is the consistent value.)

3. While you're evaluating the bond, your supervisor inquires about the relationship between time to maturity and bond Macaulay durations for:

   - A zero-coupon bond
   - A fixed-rate premium bond
   - A fixed-rate discount bond

   How would you elucidate these relationships? (6 marks)

   Zero-coupon bond. $\text{MacDur}=T$ exactly: the only cash flow arrives at maturity, so the present-value-weighted average time is the maturity itself. Its modified duration is smaller, $T/(1+y/k)$.

Fixed-rate premium bond (coupon rate above the YTM). MacDur increases as the maturity lengthens, but at a decreasing rate: it is always below T and converges to the perpetuity limit $(1 + k/y)$. At $y = 0.10$ and $k = 1$ that ceiling is 11.0 years, so extending maturity keeps adding duration but with diminishing effect.

Fixed-rate discount bond. MacDur also generally rises with maturity, but for long-term **deep-discount** bonds a duration reversal appears: MacDur rises, peaks, then declines back towards $(1 + k/y)$ as $T \to \infty$. At $y = 0.10$ a 3% coupon bond peaks at 12.68 years ($T = 30$) and falls to 11.67 years at $T = 50$; a 1% coupon bond peaks at 17.05 years. In every case MacDur $\leq T$, with equality only for a zero.

4. After assessing the bond's duration, you decide to refine your price movement estimate by considering convexity. Using the same bond details, calculate the bond's approximate convexity for a 25-bps yield change. (5 marks)
approximate convexity $\approx \frac{V_- + V_+ - 2V_0}{(\Delta y)^2 V_0}$
$= \frac{95.08108 + 93.75662 - 2(94.41595)}{(0.0025)^2(94.41595)} = \frac{0.00580}{0.00059010} \approx 9.83$
The answer is **convexity ≈ 9.8** (in the annual measure). Keep enough decimals: the numerator is a difference of numbers near 188.83, so rounding the prices to two decimals gives 0 and to four decimals gives 10 — both are pure rounding artifacts. The analytic value, the present-value-weighted second derivative of the full price at the 10% yield, is 9.83.

5. The bond position's total principal value is $100,000$. The fixed-income portfolio you're reviewing consists solely of this bond and a zero-coupon bond valued at $95,000$ with a time to maturity of 4.75 years (hint: assume annual compounding for the zero). Using the approximate modified duration from the previous question, determine the portfolio's modified duration. (5 marks)

The zero's modified duration. With annual compounding and a face value of 100,000, the market value 95,000 implies $95,000 = 100,000/(1 + y_z)^{4.75}$, hence $y_z = 0.01086$ (1.086%). Its Macaulay duration is the maturity, 4.75 years, so its modified duration is $4.75/(1 + y_z) = 4.70$ years — not 4.75.

Market-value weights (full prices). bond position: $100,000/100 \times 94.41595 = \$94,415.95$; zero: $\$95,000$; total $\$189,415.95$. Hence $w_{\text{bond}} = 49.85\%$ and $w_{\text{zero}} = 50.15\%$.
$\text{ModDur}_{\text{portfolio}} = w_{\text{bond}}\text{ModDur}_{\text{bond}} + w_{\text{zero}}\text{ModDur}_{\text{zero}} = 0.4985(2.8056) + 0.5015(4.699) \approx 3.76$ years
The weights must be **market-value** weights (weighting by principal, or by par value, has no theoretical content). Using clean prices instead changes the answer only marginally, to 3.78 years.

Problem Solving Case #2 (25 points)

You are evaluating two fixed-income securities. One is a 12.90-year maturity zero-coupon bond with a yield to maturity of 9%. This zero has modified duration of 11.83 years and convexity of 150.9. The other is a 30-year maturity, 4%-annual-payment coupon bond trading at the same yield to maturity and with the same modified duration. However, the coupon bond has higher convexity of 236.4.

1. Suppose unexpectedly positive job market data has been released. Anticipating that the central bank will adopt a more cautious approach regarding the timing of policy benchmark rate cuts, both short-term and long-term market interest rates have risen. As a result, the yields on both zero-coupon and coupon bonds increased by 1.5% to 10.50%.

   What is the actual percentage change in bond price? What is the change predicted by duration? What is the change predicted by duration plus convexity? (9 points)

There is no change to the term to maturity of either bond. Include your calculation steps and summarize your results in the table.

1. zero-coupon bond:

$P_0 = \frac{100}{(1.09)^{12.9}} = 32.90$

$P_1 = \frac{100}{(1.105)^{12.9}} = 27.58$

actual $\%\Delta P = \frac{27.58-32.90}{32.90} = -16.16\%$

2. coupon bond:

$P_0 = 4\frac{1-(1.09)^{-30}}{0.09} + 100(1.09)^{-30} = 48.63$

$P_1 = 4\frac{1-(1.105)^{-30}}{0.105} + 100(1.105)^{-30} = 41.19$

actual $\%\Delta P = \frac{41.19-48.63}{48.63} = -15.30\%$

predicted by duration: $\%\Delta P = -\text{ModD} \times \Delta y = -11.83(0.015) = -17.75\%$

predicted by duration + convexity: $\%\Delta P = -\text{ModD}\ \Delta y + \frac{1}{2}\ \text{CX}\ (\Delta y)^2$

1) zero: $-17.75\% + 0.5(150.9)(0.015)^2 = -16.05\%$

2) coupon: $-17.75\% + 0.5(236.4)(0.015)^2 = -15.09\%$

Bond	Actual % change	Predicted by duration	Predicted by duration + convexity
Zero-coupon bond (Bond A)	−16.16%	−17.75%	−16.05%
Coupon bond (Bond B)	−15.30%	−17.75%	−15.09%

2. Suppose the scenario described in question 1 did not occur. Instead, imagine there was a spike in unemployment coupled with a sharp decline in the CPI data, leading to market concerns that overly aggressive tightening has precipitated a recession. In reaction to these market conditions, the central bank announced a 50-basis-point (bps) rate cut in its latest rate decision meeting. Following this announcement, the yields on both zero-coupon and coupon bonds decreased by 1.5% to 7.5%.

   What is the actual percentage change in bond price? What is the change predicted by duration? What is the change predicted by duration plus convexity? (9 points)

1. zero-coupon bond:

$P_2 = \frac{100}{(1.075)^{12.9}} = 39.34$

actual $\%\Delta P = \frac{39.34-32.90}{32.90} = 19.57\%$

2. coupon bond:

$P_2 = 4\frac{1-(1.075)^{-30}}{0.075} + 100(1.075)^{-30} = 58.66$

actual $\%\Delta P = \frac{58.66-48.63}{48.63} = 20.63\%$

predicted by duration: $\%\Delta P = -\text{ModD} \times \Delta y = -11.83(-0.015) = 17.75\%$

predicted by duration + convexity: $\%\Delta P = -\text{ModD}\ \Delta y + \frac{1}{2}\ \text{CX}\ (\Delta y)^2$

1) zero: $17.75\% + 0.5(150.9)(0.015)^2 = 19.44\%$

2) coupon: $17.75\% + 0.5(236.4)(0.015)^2 = 20.40\%$

Bond	Actual % change	Predicted by duration	Predicted by duration + convexity
Zero-coupon bond (Bond A)	19.57%	17.75%	19.44%
Coupon bond (Bond B)	20.63%	17.75%	20.40%

3. Compare the two bonds under both market scenarios. Which bond appears more attractive? Which bond is a "long convexity" trader most likely to favor? (7 points)
   Bond B (the coupon bond) is consistently more attractive. The two bonds have the same modified duration (11.83), so the comparison is driven purely by convexity: convexity is always beneficial — it adds price when yields fall and cushions the price when yields rise. Bond B therefore loses less in the sell-off (-15.30% vs −16.16%) and gains more in the rally (20.63% vs 19.57%). Its cash flows are more dispersed (they are spread over 30 years rather than concentrated at one maturity), which is exactly what produces the higher convexity (236.4 vs 150.9). A "long convexity" trader will favor Bond B: it delivers more convexity per unit of duration risk, and here it does so at the same yield.